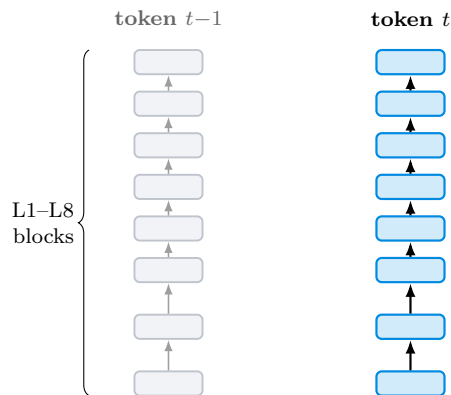


# Inference-time information flows

## Depth recurrence OFF

### Vanilla transformer

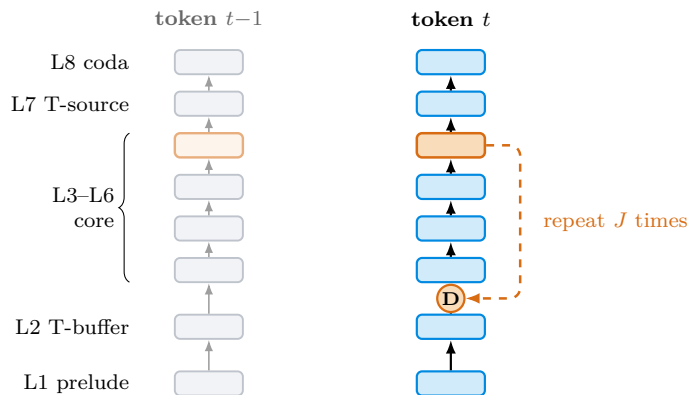
Each token runs L1–L8 once; no state is carried between tokens.



## Depth recurrence ON

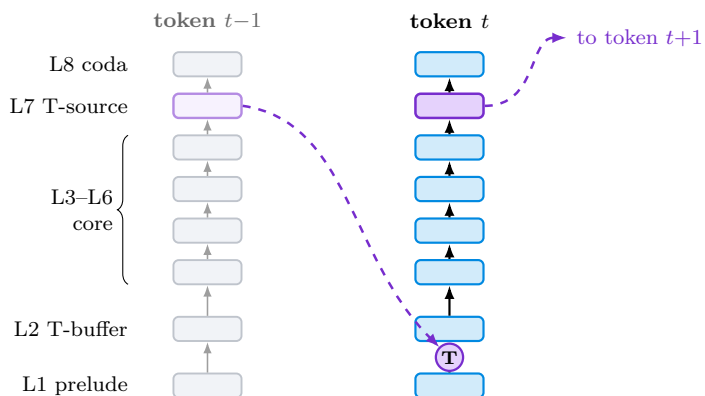
### Depth-recurrent / looped LM

The core (L3–L6) runs  $J$  times per token; its output re-enters before the core. Depth state is discarded at the next token.



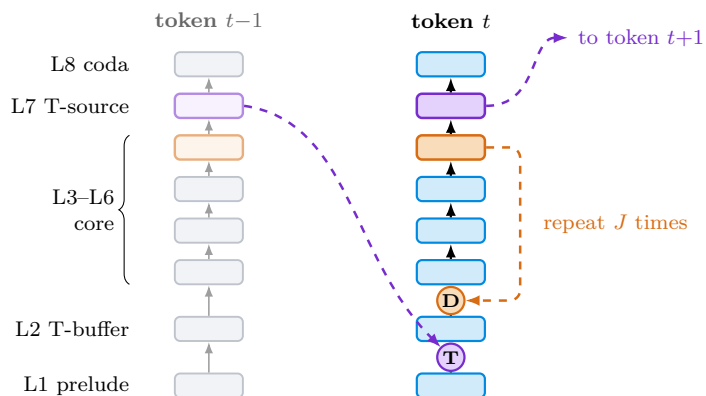
### Temporally recurrent

Token  $t-1$ 's T-source output (L7) is read by token  $t$  after the prelude, so the chain runs through every earlier token.



### Depth + temporal hybrid

Temporal state crosses tokens; depth state loops only within the current token.



token  $t-1$  layer

token  $t$  layer

writes depth state

writes temporal state

within-token flow

depth read (loop)

temporal read

state passed to next token